

Yuanyi Chen

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Personal Information

Male, 21 Ethnicity: Han Hometown: Wenzhou, Zhejiang High School: Wenzhou High School, Zhejiang Province

Education

Undergraduate **B.Eng. in Information Engineering**, *Shanghai University – Sino-European Institute of Technology*, Shanghai

Academic GPA: 3.81/4.00 (Average: 93.68/100) Rank: 10/149 (6.3%)

Key Courses Embedded System Design (90), C Programming (94), Signal Analysis & Processing A (90), Electrical Engineering & Circuits (95), Electromagnetic Field Theory A (96)

English CET-4: 575 CET-6: 577

Internship Experience

Aug – Sep 2025 **PSD Home Mobile Data Collection Software Development**

Aug – Sep 2025 **pH Measurement Software Based on HSV Color Space**

Jan – Mar 2026 **Qile You Ticketing System Based on Sunmi L3 Device**

Research Experience

Nov 2025 – Present **LunaCane: Intelligent Elderly-Assistance Perception System via Edge-Cloud Collaboration & LLM**, *Core Developer (Backend Architecture & Perception Algorithm)*

Addressing the limitations of mainstream wearable devices (e.g. smartwatches) in fall detection due to placement shift and lack of ground-interaction data, developed a non-wearable cane-based intelligent sensing system. Built an edge-cloud architecture with ESP32 and FastAPI, and led the design of the driver-to-Python-backend communication pipeline.

Jan – Mar 2025 **Idol Competition System Optimization via Differentiable State-Space Model & Bottom-K Mechanism**, *First Author*

Constructed DESSM (Differentiable Elimination State-Space Model) to infer latent approval ratings and performed counterfactual simulation to quantify rule sensitivity (finding a ~18% amplification effect of proportional fusion on fan preferences). Proposed a Bottom-K two-stage decision mechanism with utility function optimization.

Accepted by EI Conference AICBD 2026 (ID: AICBD-521557)

Dec 2024 – Nov 2025 **Multimodal Comparative Study on ASD Screening for Young Children**, *Research Assistant*

Assisted the team in constructing a mixed-scale guided gaze estimation network targeting the abnormal visual focus and appearance features exhibited by ASD children in paradigm tasks, aiming to improve the accuracy of automated screening.

Awards

- 22nd Chen Jingrun Youth Invention Award (Shanghai) **First Prize**
- 14th Shanghai University Student Engineering Practice & Innovation Competition **First Prize**
- 19th Challenge Cup Shanghai University Student Science & Technology Competition **Second Prize**

- 18th Shanghai University China International University Student Innovation Competition **Second Prize**

Social Practice

- 2024 Shanghai University “I Speak for SHU” Campaign
- 2025 Shanghai University Voluntary Blood Donation